Shoe Assembly (PU/Microfiber Upper + TPR/EVA/Rubber Outsole) – Women’s Boots

Methods

Goal and Scope

This study evaluates the climate impacts of footwear assembly at a Tier-1 supplier facility. The functional unit is defined as one pair of women’s boots, with an average weight of ~1 kg. The scope covers the main shoe assembly stages—cutting, stitching, and lasting—representing the most common production route at the facility.

In this assessment, material inputs are excluded, and the analysis focuses only on the processing stages, including energy consumption, adhesives, primers, cleaners, and waste flows. The modeling is based on a global dataset, which can be adapted in future analyses to reflect regional energy mixes or supplier-specific practices.

Data Collection and Sources

Foreground data were collected directly from the supplier through a structured LCA data template and iterative feedback exchanges. The data include measured energy use, process chemicals, and wastage rates, supported by MSDS/technical datasheets for adhesives, primers, paints, and cleaners. The refined datasets were modeled in Brightway2.5 and linked with generic background processes (e.g., electricity supply, waste treatment) from the ecoinvent database.

Impact Assessment

We quantified environmental impacts using the Environmental Footprint (EF) 3.1 method, which covers multiple midpoint impact categories (e.g., climate change, acidification, eutrophication, resource use, toxicity).

Process Descriptions

- **Cutting stage**

This dataset is expressed per pair of shoes and represents the cutting stage, the first step in the assembly sequence.

It includes electricity use and oil paint for print, with losses from off-cuts and defects incorporated into the input/output ratio of ~1.0382.

Material inputs (uppers, outsoles, etc.) are excluded here and should be modeled separately with their specific compositions.

The dataset is linked to global electricity supply (medium voltage) and waste treatment processes for textile and polyurethane residues, representing both material off-cuts and chemical waste streams.

Table 1 : inputs/outputs per pair – Cutting stage

Exchange	Amount	Unit	Location	Comment
Product output – women's boots	1.000	pair	GLO	Reference flow (functional unit)
Electricity (medium voltage)	0.020	kWh	GLO	Measured per process line/machine
Oil paint for print	0.0128	kg	GLO	Paint used in screen printing (MSDS requested for composition)
Textile waste to treatment	0.000	kg	RoW	Waste fraction included based on reported loss rates

Exchange	Amount	Unit	Location	Comment
Polyurethane waste to treatment	0.010	kg	RoW	Chemicals sent to waste

- **Stitching stage**

This dataset is expressed per pair of shoes and covers electricity use, adhesives, and chemical waste flows.
Wastage at this stage is mainly related to defective products, with an input/output ratio of ~1.006.
Material inputs (uppers, outsoles, etc.) are not included here and must be modeled separately with their specific compositions.
The dataset is linked to global electricity supply (medium voltage) and waste treatment processes for textile and polyurethane residues, representing chemicals and off-cuts sent to waste.

Table 2 : inputs/outputs per pair – Stitching stage

Exchange	Amount	Unit	Location	Comment
Product output – Women’s boots	1	pair	GLO	Reference flow (functional unit)
Electricity (medium voltage)	0.1	kWh	GLO	Measured – direct consumption per process line/machine
Stitching glue (163N)	0.0006	kg	GLO	Placeholder for glue (MSDS provided separately)
Textile waste	0	kg	RoW	Included based on loss rates (no additional losses reported here)
Polyurethane waste	-0.01	kg	RoW	Chemicals sent to waste

- **Lasting stage**

This dataset is expressed per pair of shoes and represents the lasting stage, the final step in the assembly sequence.

It includes electricity use, polyurethane adhesive, primer (BW553), and a shoe surface cleaner, reflecting typical material and chemical requirements at this stage.

Reported wastage is mainly due to defective products, with an input/output ratio of ~1.004.

Material inputs for uppers and outsoles are excluded and modeled separately with their specific compositions.

The dataset is linked to global electricity supply (medium voltage) and waste treatment processes for polyurethane residues, representing chemicals sent to waste.

Table 3 : inputs/outputs per pair – Lasting stage

Exchange	Amount	Unit	Location	Comment
Product output – women's boots	1.000	pair	GLO	Reference flow (functional unit)
Electricity (medium voltage)	1.200	kWh	GLO	Measured per process line/machine
Polyurethane adhesive (water-based)	0.020	kg	GLO	Adhesive use, per supplier data (MSDS provided)
Primer (BW553)	0.020	kg	GLO	Primer for shoes
Shoe surface cleaner	0.006	kg	GLO	Cleaner applied in finishing step
Polyurethane waste to treatment	0.010	kg	RoW	Chemicals sent to waste

- **Shoe Assembly**

This dataset represents the combined shoe assembly stage, built by linking the three main sub-processes — cutting, stitching, and lasting.

Across all stages, the cumulated material loss is ~4.8%, consisting of:

- Cutting off-cuts,
- Defective products at each stage,
- Final rejects.

Waste fractions (e.g., polyurethane and textile scraps) were modeled explicitly and linked to a generic textile waste treatment process.

Table 4 : inputs/outputs per pair – Shoe Assembly

Exchange	Amount	Unit	Location	Comment
Product output – women's boots	1.000	unit	GLO	Reference flow (functional unit)
Lasting stage (aggregated input)	1.000	unit	GLO	Lasting is the final step

Exchange	Amount	Unit	Location	Comment
Stitching stage (aggregated input)	1.004	unit	GLO	Considering loss rate lasting
Cutting stage (aggregated input)	1.010	unit	GLO	Considering loss rate lasting and stiching
Textile waste to treatment	0.0487	kg	RoW	Cumulative material losses across cutting, stitching, and lasting (~4.7%)

Life Cycle Impact Assessment (LCIA) Results

Table 5 presents the results of the life cycle impact assessment for the shoe assembly stage, expressed per pair of women's boots (functional unit). The EF 3.1 midpoint method was applied, covering a wide range of environmental issues including climate change, resource use, ecotoxicity, land use, and human toxicity. To put the supplier-specific dataset into perspective, we compared it against generic assembly datasets available in EF 3.1 for open-toed, close-toed, and boot categories.

Table 5 : EF 3.1 Impact Results (per pair of shoes, functional unit = 1 pair)

Method	Unit	Supplier – Women's Boots	Generic Open-Toed Shoes - EF3.1	Generic Close-Toed Shoes - EF3.1	Generic Boots - EF3.1
Acidification (ACD)	mol H^+ eq/pair	0.0060	0.0132	0.0142	0.0145
Ecotoxicity, freshwater (ETF)	CTUe/pair	10.47	52.86	56.48	56.81
Resource use, fossils (FRU)	MJ/pair	17.46	29.08	31.66	32.38
Eutrophication, freshwater (FWE)	kg P eq/pair	0.00048	2.68e-5	3.61e-5	3.82e-5
Climate change (GHG)	kg CO_2 eq/pair	1.28	2.17	2.32	2.36
Human toxicity, cancer (HTC)	CTUh (can.)/pair	3.93e-9	6.12e-9	6.80e-9	6.82e-9
Human toxicity, non-cancer (HTN)	CTUh (non-can.)/pair	9.86e-9	3.98e-8	4.14e-8	4.17e-8

Method	Unit	Supplier – Women's Boots	Generic Open-Toed Shoes - EF3.1	Generic Close-Toed Shoes - EF3.1	Generic Boots - EF3.1
Ionizing radiation (IOR)	kBq U^{235} eq/pair	0.143	0.115	0.125	0.128
Land use (LDU)	Pt/pair	2.70	7.65	8.07	8.15
Resource use, minerals/metals (MRU)	kg Sb eq/pair	3.74e-6	4.37e-6	5.32e-6	5.52e-6
Ozone depletion (OZD)	kg CFC-11 eq/pair	1.57e-8	1.85e-8	2.35e-8	2.36e-8
Photochemical ozone formation (PCO)	kg NMVOC eq/pair	0.00415	0.00912	0.00978	0.00993
Particulate matter (PMA)	disease inc./pair	6.80e-8	4.25e-7	4.44e-7	4.49e-7
Eutrophication, marine (SWE)	kg N eq/pair	0.00169	0.00239	0.00262	0.00267
Eutrophication, terrestrial (TRE)	mol N eq/pair	0.0128	0.0242	0.0264	0.0269
Water use (WTU)	m^3 eq/pair	0.289	0.889	1.031	1.067

Overall, the supplier dataset shows consistently lower impacts across most categories. For instance, climate change impacts amount to 1.28 kg CO_2 eq/pair, which is roughly 45% lower than the generic boots dataset (2.36 kg CO_2 eq/pair). Resource use, fossils is also considerably reduced (17.5 MJ/pair compared to 29–32 MJ/pair in the generic datasets). In contrast, categories such as freshwater ecotoxicity and land use are significantly higher in the generic datasets, likely reflecting modeled upstream assumptions and less refined treatment of waste and chemical flows. This comparison highlights the value of primary supplier data, which yields more representative and often lower impact profiles than reliance on generic databases alone.

Contribution Analysis of Climate Change Impacts

To better understand the drivers of climate change impacts within the assembly stage, we conducted a contribution analysis at two levels. First, the **aggregate shoe assembly process** is assessed, covering cutting, stitching, lasting, and waste treatment presented at figure 1

Second, we examine the **lasting step in detail**, since it was identified as the most impactful stage in the assembly process and is presented at figure 2. The treemaps presented below show the relative contribution of individual steps and inputs, highlighting the main hotspots for targeted improvement.

Figure 1: Assembly Impact Treemap

The overall shoe assembly stage is dominated by the **lasting step**, which contributes ~85% of the climate change impact. Stitching and cutting are comparatively minor (around 6% each), while waste treatment accounts for ~4%. This reflects the fact that the lasting stage involves multiple energy- and chemical-intensive operations (adhesives, primers, cleaners), whereas cutting and stitching mainly rely on manual or lower-energy processes.

Waste treatment remains relatively minor compared to in-process energy and chemical inputs due to the low loss rates observed at each assembly steps.

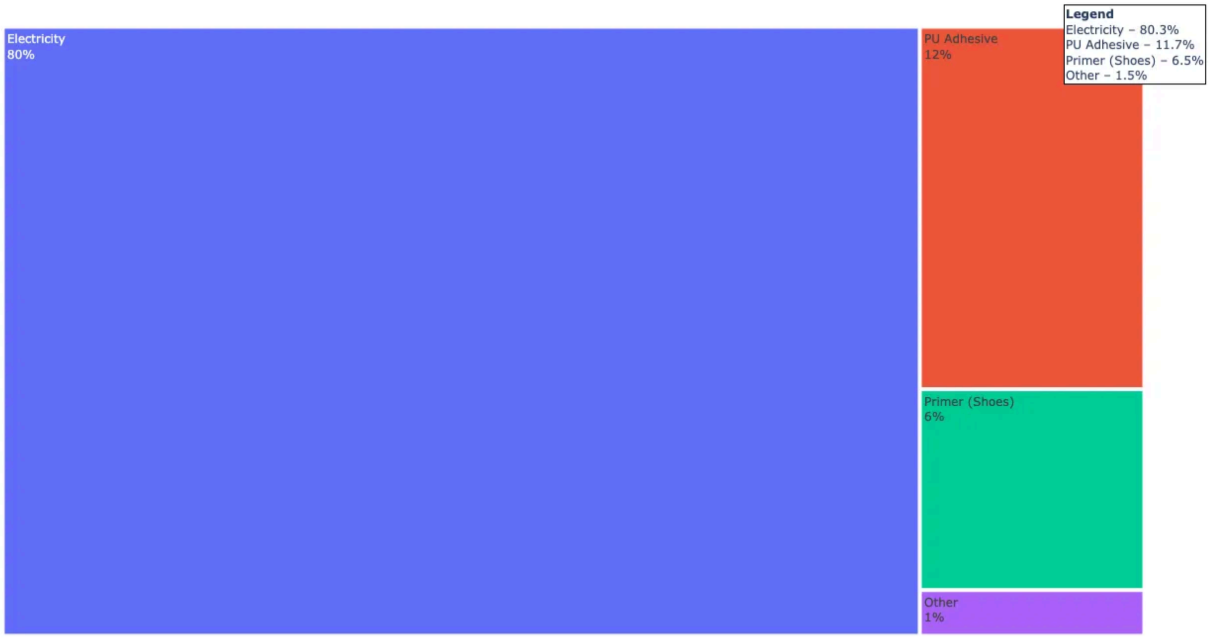

Figure 2: Lasting Impact Treemap

At the level of the lasting process itself, the picture becomes clearer. Electricity consumption dominates (80%), followed by PU adhesives (~12%) and primers (~6%). Cleaners and minor inputs are grouped into "Other" (~1.5%).

Electricity is the main driver of impacts in lasting, emphasizing the importance of grid decarbonization or efficiency improvements (e.g., optimized heating/curing equipment).

PU adhesives and primers are the key chemical contributors, together accounting for almost 20% of the impact. Cleaner use is minor, but relevant given chemical exposure risks.

Appendix

The life cycle inventory is available in the file below

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